# Alexander **Zhuk**

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## Education

**Illinois Institute of Technology** *Chicago, IL*

B.S. in Civil Engineering GPA 4.0 *Expected May 2029*

## Skills

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|---|---|
| **CAD** | SolidWorks \| AutoCAD \| Inventor \| Revit \| Blender |
| **Software** | Ansys Fluent \| Visual Studio Code \| MATLAB \| Unity |
| **Fabrication** | Lathe Machining \| Mill Machining \| Soldering \| Metal working |
| **Digital Media** | Photoshop \| Lightroom \| Microsoft 365 |
| **Languages** | English \| Ukrainian \| Russian |

## Experience

**IIT Motorsports (Formula SAE)** *Chicago, IL*

Vice President & High Voltage Team Lead *August 2025 to Present*

- Directed workflow and project timelines for a 50-member team Formula SAE team, establishing a comprehensive 2-year vehicle build schedule.
- Led a 6-person specialized sub-team in the end-to-end design and mechanical integration of a 2026 electric tractive battery system.

**AZcore (FSAE)** *Chicago, IL*

LARP (Lithium Array Rechargeable Pack) *August 2026 to Present*

- Engineered a modular multi-plate battery assembly utilizing standardized commercial off-the-shelf components, streamlining machining and assembly cycles while increasing mechanical performance compared to previous iterations.
- Optimized battery module architecture using SolidWorks, successfully increasing pack density by 6% and overall performance by 4%.
- Developed and validated a custom active cooling system for a 72s6p accumulator module using ANSYS Fluent Conjugate Heat Transfer (CHT) simulations, reducing operating temperatures by 5°C and boosting thermal efficiency by 7%.

**American Society of Civil Engineers Steel Bridge** *Chicago, IL*

Fabrication Team Member *August 2025 to Present*

- Manufactured structural steel components for the competition bridge, utilizing shop tools and mill machinery to meet precise design specifications.
- Collaborated with 6 team leads to review bridge design to ensure all components were properly drafted.

**Structural Engineers Association of Illinois** *Chicago, IL*

VP of Membership *December 2025 to Present*

- Drafted a comprehensive 2026-2027 events calendar, including technical workshops, site visits, and structural design competitions to increase member engagement.

**Plainfield North Theater** *Plainfield, IL*

Lead Structural Designer & Sound Tech Lead *August 2023 to May 2025*

- Directed structural design and construction of multi-floored theatrical sets to safely support a 30+ member cast
- Managed live signal levels and channel activation for 20+ wireless microphones, ensuring precise audio cues and consistent gain staging throughout multi-act performances

## Involvement

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|---|---|
| **Project Teams** | IIT Motorsports (FSAE) \| Steel Bridge |
| **Prof. Societies** | SEAOI \| ASCE \| SHPE |
| **Campus Org** | Delta Tau Delta (ΔΤΔ) \| Hawk Ambassador \| Exelon Summer Institute |
| **Media** | Photography Club |